



Configured drive

Motor - DC-max22S GB KL 12V
Gearbox - GPX22 C 35:1
Sensor - ENX10 EASY 1024IMP

Item number: B82D23A9690C Revision number 1

General Terms and Conditions: https://www.maxongroup.ch/maxon/view/content/terms_and_conditions_page

The STEP file is then available via the symbol.

B82D23A9690C.stp (STP AP 214)

Open configuration: <https://www.maxongroup.ch/maxon/view/configurator/?ConfigID=B82D23A9690C>

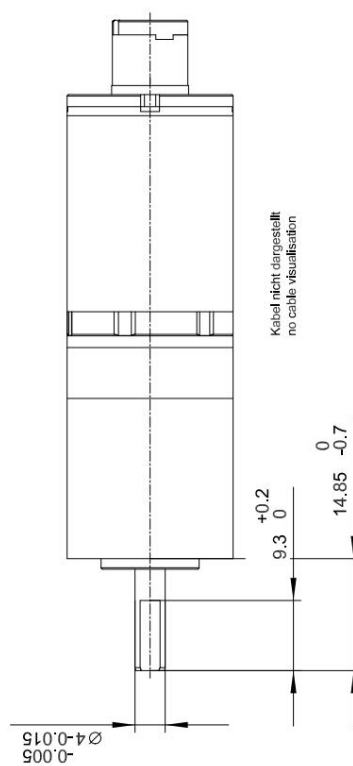
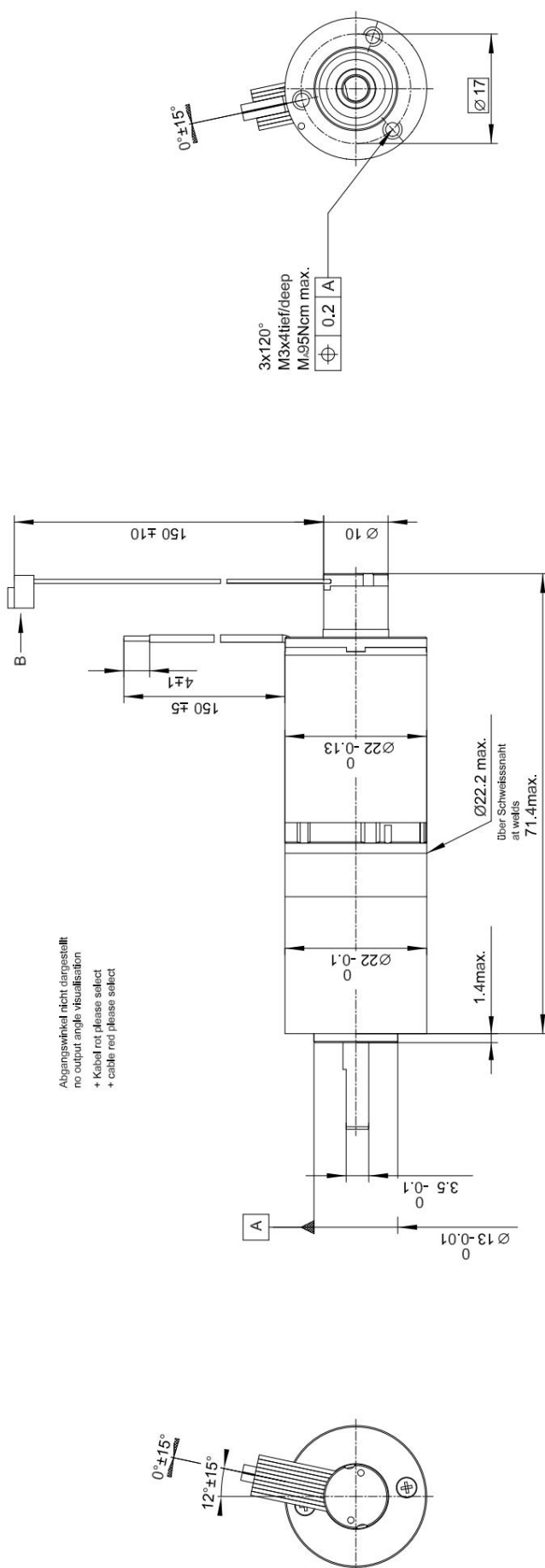
Gearbox axial play: 0...0.2mm

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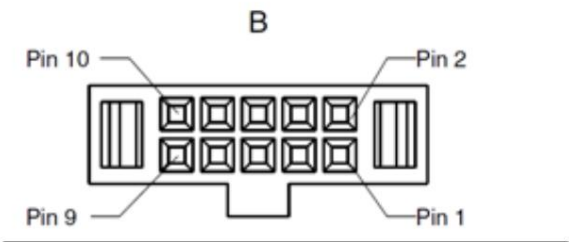
Drawing not to scale!



Kabel nicht dargestellt
no cable visualisation



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Connector type encoder

1.27mm 10-pin

- Pin 1 Do not connect
- Pin 2 VCC Pin 3
- GND
- Pin 4 Do not connect Pin 5
- Channel A \
- Pin 6 Channel A
- Pin 7 Channel B \
- Pin 8 Channel B
- Pin 9 Channel I \
- Pin 10 Channel I



Summary of your selected configuration

Total weight of the drive: 110 g

DC-max22S GB KL 12V

Product details	
Commutation	Graphite brushes
Nominal voltage	12
Engine bearing	Rolling bearings
Electrical connection motor	
Electrical connection motor	Cable
Cable length	150

GPX22 C 35:1

Product details	
Gearbox type	CA
Reduction	35
Number of levels	2

ENX10 EASY 1024IMP

Product details	
Pulse count	1024
Connector type encoder	1.27mm 10-pin

Legend for article description

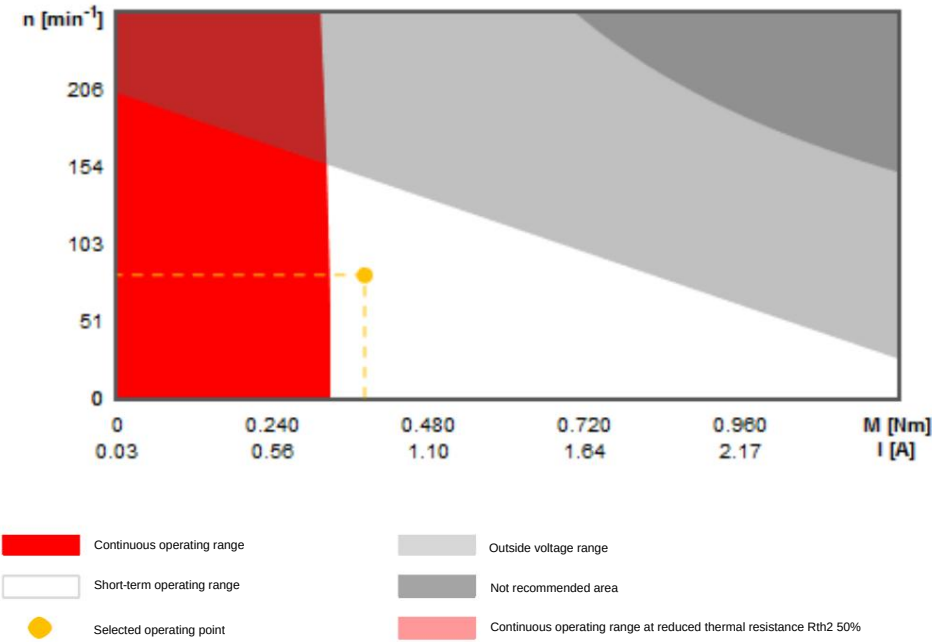
EC motor:		DC motor:		IDX Drive/Motor:	
A	Version with Hall sensors	CLL	Capacitor long life (spark suppression)	AWAY	Integrated brake
B	Sensorless design	DC-max	DC motor (brushed)	CO	CANopen interface
BL	Brushless	DCX	DC motor (brushed)	ENC	Encoder
C	Integrated electronics	EB	Precious metal brushes	ET	EtherCAT interface
ECX	EC motor (brushless)	GB	Graphite brushes	IDX	Drive system
FL	FLAT (type series name)	KL	ball bearings	IO	I/O interface
HP	High Power	S/M/L	Short / Medium / Long	P	Positioning control
HTQ	High Torque	SL	Sintered plain bearings	S	Speed control
HTQF	High Torque, fan-ventilated			S/M/L	Short / Medium / Long
KL	ball bearings				
M/L/XL	Medium / Long / Extra Long				
PR	PRIME (type series name)				
SP	SPEED (type series name)				
HOURS	standard				
STE	Sterilizable				
STEC	Sterilizable, ceramic bearings				
TQ	TORQUE (type series name)				

Encoder/Sensor:	
ABS	absolute
BiSSC	Bidirectional serial synchronous (protocol)
COMM	commutation signals
EASY	(Type series name)
EMT	(Type series name)
ENX	Encoder
FFC	Flex cable
GAMA	(type series name)
IMP	Impulses
INT	Integrated
MAG	(Type series name)
QUAD	(type series name)
RIO	(Type series name)
SSI	Synchronous serial interface (protocol)
XT	Extended version

Gearbox	
A	Standard version
C	Ceramic version
GPX	planetary gear
HP	High Power
LN	Noise reduced
LZ	Reduced play
SP	SPEED (type series name)
ST	Number of levels
STE	Sterilizable
STES	Sterilizable, with seal
UP	Ultra Performance



Drive design



Combination details

Your entries

Available voltage	12 V
speed	82 min-1
torque	0.38 Nm

Maximum values of the drive at available voltage

Available voltage	12 V
Max. speed at your load	146 min-1
Max. continuous torque	0.35 Nm
Max. continuous load current	0.82 A

Electrical values for your operating point

speed	82 min-1
torque	0.38 Nm
Required voltage	8.17 V
Required power	0.88 A



DC-max22S GB KL 12V

Product specification		
Values at nominal voltage		
	Nominal voltage	12 V
	Idle speed	7140 rpm
	No-load current	28.3mA
	Rated speed	5260 rpm
	Rated torque (max. continuous torque)	12.5 mNm
	Rated current (max. continuous load current)	0.819 A
	Stopping torque	49.4 mNm
	Starting current	3.14 A
	Max. efficiency	80.6%
Key data		
	Max. continuous power output	10.4 W
	Connection resistance	3.83 μ
	Connection inductance	0.274 mH
	Torque constant	15.8 mNm A-1
	Speed constant	606 min-1 V-1
	Characteristic curve slope	147 min-1 mNm-1
	Mechanical start-up time constant	8.57ms
	Rotor moment of inertia	5.57 gcm2
Thermal data		
	Thermal resistance housing-air	20 KW-1
	Thermal resistance winding-housing	6 KW-1
	Thermal time constant of the winding	17.2 s
	Thermal Time constant of the motor	538 s
	Ambient temperature	-30...85 °C
	Max. winding temperature	125 °C
Mechanical data		
	Limit speed	9000 rpm
	Axial play	0.05...0.15 mm
	Preload Radial	0 N
	clearance	0.025 mm
	Max. axial load (dynamic)	3.3 N
	Max. axial pressing force (static)	45 N
	static, shaft supported Max.	420 N
	radial load 5 mm from flange Measurement from	12.3 N
	flange	5 mm
Further specifications		
	Number of pole pairs	1
	Number of collector segments	9



Engine weight	46.9 g
Engine length	33.2 mm
Typical noise level	55 dBA (6000 min-1)

More information about the motor: <https://www.maxongroup.com/maxon/view/product/DC-MAX22S01GBKL598?download=show>



GPX22 C 35:1

Product specification

Transmission data

Reduction	35:1
Absolute reduction	4554/130
Number of levels	2
Max. continuous torque	0.90 Nm
Short-term permissible torque	1.2 Nm
Direction of rotation, input to output	=
Max. efficiency	81%
Average gear play unloaded	1.1 °
Mass moment of inertia Max.	0.501 gcm ²
transmittable power continuously	15 W
Max. transferable power short-term	19 W

Technical data

Output shaft bearing	Rolling bearings
Max. radial clearance, 5 mm from flange	max. 0.2 mm
Axial play	0...0.2 mm
Max. permissible radial load, 5 mm from flange Max.	100 N
permissible axial load	40 N
Max. permissible pressing force	100 N
Max. input speed continuous Max. input	10000 rpm
speed short-term Recommended	12500 rpm
temperature range	-40..100 °C

More information about the gearhead: <https://www.maxongroup.com/maxon/view/product/GPX22CAKLSL35D0CPLW?download=show>



ENX10 EASY 1024IMP

Product specification		
Sensor data		
	Number of pulses per revolution	1024
	Number of channels	3
	Line Driver	RS422
	Max. electrical speed	90000 rpm
	Max. mechanical speed	30000 rpm
Technical data		
	Supply voltage Vcc	5 V ±10%
	Output signal	INC
	Output signal driver	Differential / EIA RS 422
	Output current per channel	-20...20 mA
	State length	20...160 °C
	Signal rise time/signal fall time	20/20 ns
	Min. state duration	125 ns
	Direction of rotation	A before B CW
	Index position	A low and B low
	Index synchronous to AB	Yes
	Index pulse width	90 °el
	Typical power consumption at standstill	22 mA
	Max. moment of inertia of the pulse disc	0.05 gcm2
	Operating temperature range	-40...100°C
	Number of sterilization cycles	0

More information about the encoder: <https://www.maxongroup.com/maxon/view/product/ENX10EASY06?download=show>